

Fourier Series + Polynomial

lunedì 13 aprile 2026 10:53

Bauer 2025:

- Model based on a position-dependent Fourier series

$$\hat{\theta}_{TE} = \sum_k A_k \cdot \sin(f_k \cdot \theta + \varphi_k) + a_0$$

- State-dependent coefficients A_k , φ_k and a_0

$$\begin{aligned} A_k(\tau_{load}, \dot{\theta}, T) = & p_{A,k,1} \cdot \tau_{load}^2 + p_{A,k,2} \cdot \dot{\theta}^2 + p_{A,k,3} \cdot T^2 \\ & + p_{A,k,4} \cdot \tau_{load} + p_{A,k,5} \cdot \dot{\theta} + p_{A,k,6} \cdot T \\ & + p_{A,k,7} \cdot \tau_{load} \cdot \dot{\theta} + p_{A,k,8} \cdot \tau_{load} \cdot T \\ & + p_{A,k,9} \cdot \dot{\theta} \cdot T + p_{A,k,10} \end{aligned}$$

- Different parameter sets for positive and negative movements

$\hat{\theta}_{TE}$	– Estimated transmission error
A_k	– State-dependent amplitude
φ_k	– State-dependent phase
a_0	– State-dependent offset
f_k	– Normalized spatial frequency
k	– Index of frequency
θ	– Motor-side position
τ_{load}	– Load torque
$\dot{\theta}$	– Motor-side velocity
T	– Temperature
$p_{,k,1...10}$	– Polynomial parameters

Signal processing Bauer 2025:

- Signal processing before Fourier series calculation:
 - Nonuniform distribution of measurement points in the position / spatial domain
 - Linear interpolation = resampling with equidistant sampling in the spatial domain = theoretically necessary condition for FFT
 - Removal of linear trends
 - Hamming window
 - Zero-padding to the next higher power of 2 above 10 times the original length
 - Two-sided amplitude spectrum
 - Window normalization
 - Single-sided spectrum
- Amplitude directly derived from single-sided spectrum
- Phase derived from equivalent sine-cosine form
- Constant offset = mean value of TE data
- Polynomial parameter optimization
 - Removal of mean
 - Standardization
 - Least Squares Optimization
 - Degree of polynomial not optimized

Signal processing Bilancia 2025:

- Signal processing before Fourier series calculation:
 - Uniform distribution of measurement points in the time domain
 - Removal of linear trends
 - Two-sided amplitude spectrum
 - Single-sided spectrum
- Amplitude directly derived from single-sided spectrum
- Phase directly derived from single-sided spectrum

Previous steps:

- Compensation results from paper Bilancia 2025 reproduced
- Pipeline for previous paper adapted to XiLab paper
- Included the calculation based on load-side angle
- Included signal processing approach from XiLab paper
- checked doubled files:
 - '200.0rpm0.0Nm25.0deg1.csv'
 - '200.0rpm100.0Nm25.0deg1.csv'
 - '800.0rpm200.0Nm25.0deg.csv' --> '800.0rpm200.0Nm25.0deg_1.csv'
 - '1100.0rpm100.0Nm30.0deg_collegamento.csv'
 - '1600.0rpm100.0Nm30.0degCollegamento.csv'
 - '1600.0rpm100.0Nm30.0degcollegamento2.csv'

Next steps:

- Optimization of degree of polynomial used (regularization + sensitivity analysis of higher order polynomial (e.g. degree 5)), exp, fmincon / SQP
- KPIs of reference Fourier Series
 - Signal processing time- (XiLab vs. improved implementation) vs. spatial-based
 - Usage of input vs. output angle
- KPIs of Fourier Series + Polynomial approach

- Benchmarks:
 - Reference Fourier Series
 - Test data
- Implementations
 - Signal processing time- (XiLab vs. improved implementation) vs. spatial-based
 - Usage of input vs. output angle
- Visual analysis of plots
 - On selected plots from previous paper (identification data)
 - On 3 test cases presented

Considerations:

- Limitation: Dataset lacks data with same sign of velocity and torque
- Measurement plan:
 - For Fourier Series full factorial measurement plan most suitable
 - For ML random distribution like sobol set better
 - For test set random distribution desirable

Preliminary results (24.04.2026):

- Signal processing approaches produce similar results
 - Results from Bauer 2025 approach seemed visually slightly better, but needs to be evaluated
- Results for input-side slightly worse, as expected due to less direct connection to cycloidal stage
 - Still desirable to use because implementation in the robot is only feasible based on input-side (without secondary encoder)
- Results with polynomial so far below expectations (Results in Bauer 2025 factor 10 better)
 - Adaption of code to XiLab dataset needs to be check again
 - Polynomial degree / structure needs to be optimized

Case	Direction	Input Definition	MSE (rad ²)	MSE (deg ²)	RMSE (rad)	RMSE (deg)	MAE (rad)	MAE (deg)	Max Error (rad)	Max Error (deg)
1	Positive	TE over input-side angle	5.6990e-08	1.871e-04	2.2683e-04	0.0130	1.8561e-04	0.0106	0.0011	0.0630
2	Positive	TE over load-side angle	3.2746e-08	1.075e-04	1.7439e-04	0.0100	1.4516e-04	0.0083	0.0011	0.0630
3	Negative	TE over input-side angle	1.8301e-07	6.01e-04	4.0868e-04	0.0234	3.5247e-04	0.0202	0.0018	0.1031
4	Negative	TE over load-side angle	1.6691e-07	5.48e-04	3.8936e-04	0.0223	3.4449e-04	0.0198	0.0018	0.1031



